Anova Homework

Question 1:“Is there a significant difference on the average textbook cost when grouped by Field of Study? Use α=0.05.”

Hypothesize-

H0: avg. textbook cost of arts = avg. textbook cost of humanities = avg. textbook cost of natural science = avg. textbook cost of social science

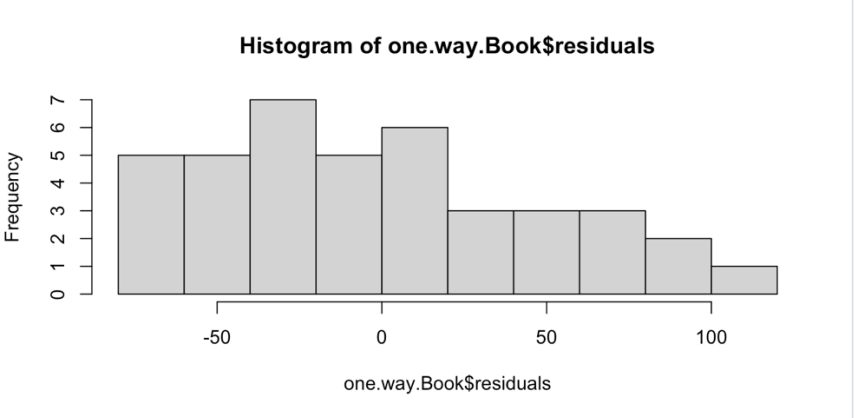
HA: at least one differs

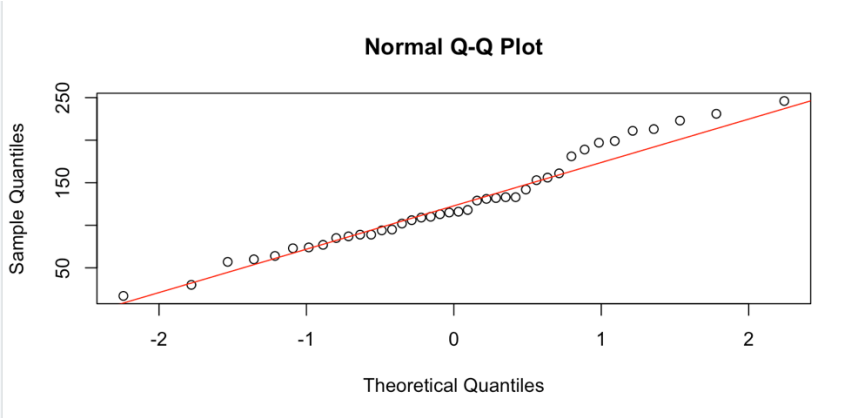
Prepare- (F-test statistic)

Random & Independent(assumed)

Normality…

(histogram and qqplot)





Graphs interpretation: Based off the histogram and the qq plot normality can be assumed.

Homoscedasticity (Equality Variances)-

Results: FALSE

Compute & Compare(AOV & Tukey)-

Results of AOV: P-Value = 0.014 (reject null hypothesis)

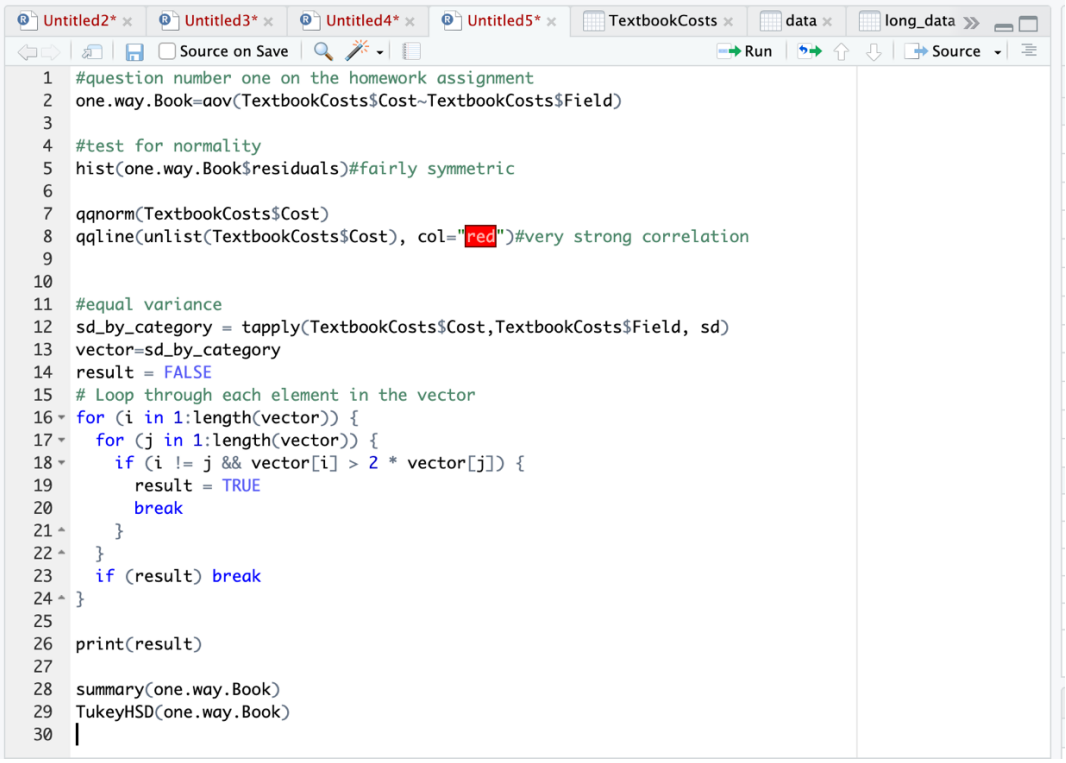
Results of Tukey: Field of study books with a statistically significant difference “Natural Science – Arts” P-Value = 0.009

Summarize results-

An analysis of variance (ANOVA) was conducted to determine whether the mean textbook cost differs across fields of study. The results of the test produced a p-value of 0.014, which is less than the chosen significance level of $\alpha = 0.05$. Therefore, we reject the null hypothesis and conclude that there is a statistically significant difference in average textbook costs among at least one pair of fields of study.

To identify which specific groups differ, a Tukey's Honestly Significant Difference (Tukey HSD) post hoc test was performed. The results indicate that the difference in mean textbook cost between the Natural Sciences and Arts fields is statistically significant ($p = 0.009$). Among all pairwise comparisons, this difference was the most statistically significant observed.

Appendix of Rstudio Code Q1:



Untitled2* | Untitled3* | Untitled4* | Untitled5* | TextbookCosts | data | long_data

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```
#question number one on the homework assignment
one.way.Book=aov(TextbookCosts$Cost~TextbookCosts$Field)

#test for normality
hist(one.way.Book$residuals)#fairly symmetric

qqnorm(TextbookCosts$Cost)
qqline(unlist(TextbookCosts$Cost), col="red")#very strong correlation


#equal variance
sd_by_category = tapply(TextbookCosts$Cost,TextbookCosts$Field, sd)
vector=sd_by_category
result = FALSE
# Loop through each element in the vector
for (i in 1:length(vector)) {
  for (j in 1:length(vector)) {
    if (i != j && vector[i] > 2 * vector[j]) {
      result = TRUE
      break
    }
  }
  if (result) break
}

print(result)

summary(one.way.Book)
TukeyHSD(one.way.Book)
```

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```
> summary(one.way.Book)
                   Df Sum Sq Mean Sq F value Pr(>F)
TextbookCosts$Field  3  30848   10283   4.055  0.014 *
Residuals           36  91294    2536
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> TukeyHSD(one.way.Book)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = TextbookCosts$Cost ~ TextbookCosts$Field)

$`TextbookCosts$Field`
                              diff        lwr        upr     p adj
Humanities-Arts               25.7  -34.95384  86.353844 0.6669143
NaturalScience-Arts           76.2   15.54616 136.853844 0.0090147
SocialScience-Arts            23.7  -36.95384  84.353844 0.7201024
NaturalScience-Humanities     50.5  -10.15384 111.153844 0.1312366
SocialScience-Humanities      -2.0  -62.65384  58.653844 0.9997441
SocialScience-NaturalScience -52.5 -113.15384   8.153844 0.1097759
```

Question 2:

“Can the paramedics conclude that the average number of calls is different between shifts? Use a 5% level of significance.”

Hypothesize-

H0: Avg. number Of calls for morning shift = Avg. number Of calls for afternoon shift = Avg. number Of calls for night shift

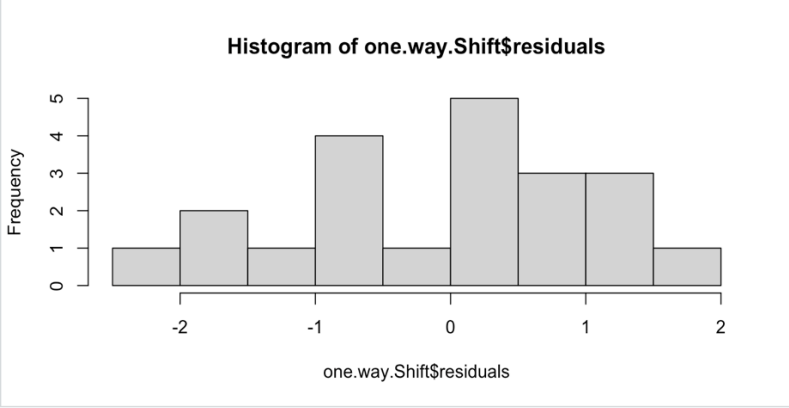
HA: At least one differs.

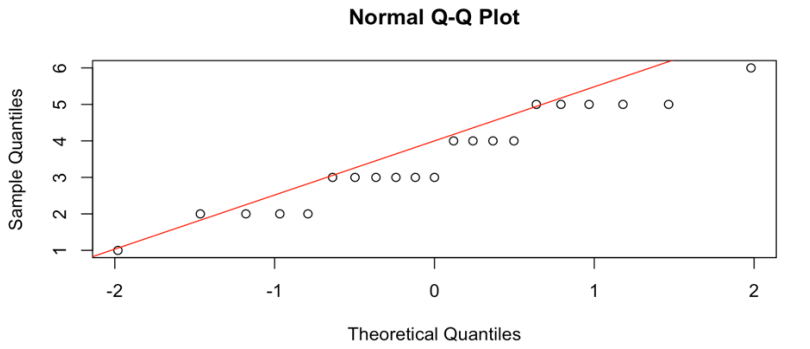
Prepare- (F-test statistic)

Random & Independent(assumed)

Normality…

(histogram and qqplot)





Graphs interpretation: Considering we're working with integer data, based off the histogram and the qq plot normality can be assumed.

Homoscedasticity (Equality Variances)-

Results: FALSE

Compute & Compare(AOV & Tukey)-

Results of AOV: P-Value = 0.039 (reject null hypothesis)

Results of Tukey: The number of calls received between ***night shift*** and ***morning shift*** yielded statistically significant differences with a p-value of 0.034

Summarize results-

To examine whether the average number of calls varies by work shift, an ANOVA test was performed. The resulting p-value of 0.039 falls below the significance threshold of 0.05, indicating that there are

statistically significant differences in call volume between at least some of the shifts.

To further explore where these differences occur, a Tukey HSD post hoc analysis was conducted. This test revealed that the night and morning shifts differ significantly in their mean number of calls ($p = 0.034$), suggesting that call volume is not consistent between these two time periods.

Appendix of Rstudio Code Q2:

```r
install.packages("tidyr")
library(tidyr)
#question number two homework

#importing data
ShiftData=data.frame(Morning=c(2,3,2,4,3,1,3), Afternoon=c(3,4,4,5,3,2,5), Night=c(5,4,5,3,2,5,6))

# Let's create the "long data", with two columns: 1) Shift; 2) Calls.
long_shiftdata <- pivot_longer(ShiftData, cols = everything(), names_to = "Shift", values_to = "num of Calls")
long_shiftdata #prints out the long data (if you want to see it)

#AOV steps...
one.way.Shift=aov(long_shiftdata$`num of Calls`~long_shiftdata$Shift)

#checking for normality
hist(one.way.Shift$residuals) #not clearly symmetric enough

qqnorm(long_shiftdata$`num of Calls`)
qqline(long_shiftdata$`num of Calls`, col='red') #adds the line


#equal variance
sd_by_category = tapply(long_shiftdata$`num of Calls`,long_shiftdata$Shift, sd)
vector=sd_by_category
result = FALSE
# Loop through each element in the vector
for (i in 1:length(vector)) {
  for (j in 1:length(vector)) {
    if (i != j && vector[i] > 2 * vector[j]) {
      result = TRUE
      break
    }
  }
  if (result) break
}
#finding results
print(result)


#find the pval
summary(one.way.Shift)
TukeyHSD(one.way.Shift)
```

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```
> #find the pval
> summary(one.way.Shift)
                    Df Sum Sq Mean Sq F value Pr(>F)
long_shiftdata$Shift  2  10.67   5.333   3.907  0.039 *
Residuals            18  24.57   1.365
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> TukeyHSD(one.way.Shift)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = long_shiftdata$`num of Calls` ~ long_shiftdata$Shift)

$`long_shiftdata$Shift`
                        diff       lwr       upr     p adj
Morning-Afternoon -1.1428571 -2.736729 0.4510146 0.1882813
Night-Afternoon    0.5714286 -1.022443 2.1653003 0.6380620
Night-Morning      1.7142857  0.120414 3.3081574 0.0339179

>
```